

Enhanced Corneal Endothelial Cell Segmentation via Frequency-Selected Residual Fourier Diffusion Models

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Abstract—Segmenting corneal endothelial cells in conditions like Fuchs endothelial dystrophy (FED) is challenging due to guttae obscuring cell details and complicating imaging. This is further compounded by labor-intensive manual annotations and a lack of large annotated datasets. To address these issues, we introduce a novel two-stage framework using Denoising Diffusion Probabilistic Models (DDPMs) for generating training pairs of corneal endothelial cell images. In the first stage, we generate synthetic endothelial labels, which are then used to guide the production of high-resolution corneal images in the second stage. We also present the Fourier Residual Block with Frequency Selection (FRB-FS), which enhances important high-frequency details for clearer textures and edges, while suppressing irrelevant low-frequency components. This is the first application of diffusion models to corneal endothelial cell segmentation. Extensive experiments and ablation studies on two benchmark datasets demonstrate the effectiveness of our framework.

Index Terms—Medical Image Analysis, Denoising Diffusion Probabilistic Model, Semantic Segmentation

I. INTRODUCTION

Accurate segmentation of medical images plays a critical role in disease diagnosis and treatment monitoring but remains challenging due to the complexity of anatomical structures and significant variations in imaging conditions [1]–[3]. In the case of Fuchs Endothelial Dystrophy (FED), the presence of guttae further complicates the segmentation of the corneal endothelium and significantly impacts overall accuracy [4]. Guttae, collagen deposits resembling droplets on Descemet’s membrane, obscure cellular features and reduce the efficacy of traditional segmentation approaches [5]–[7]. Automated segmentation methods often lack reliability, especially with healthy corneal images, and the scarcity of annotated datasets hinders training robust models.

With advancements in computer vision, automatic segmentation algorithms for corneal endothelial cells have become increasingly important, aiming to replace labor-intensive manual methods [8]–[10]. Traditional approaches that rely on frequency and morphological analysis often struggle under complex conditions, such as those presented by Fuchs endothelial dystrophy [11], [12]. Although deep learning models, such as U-Net [13] and recent structural adaptations [14], show great promise, their effectiveness is significantly limited by

the need for large amounts of accurately labeled training data. To mitigate this issue, recent studies have explored the use of generative adversarial networks (GANs) to synthesize realistic corneal endothelium images, thereby artificially expanding available training datasets [15]. Despite their potential to transform the field, GAN-based methods still face critical challenges, including pronounced image noise, substantial training instability (such as non-convergence), and the inability to generate large quantities of high-quality, labeled data [16]–[18].

In this paper, we present a novel framework that leverages Denoising Diffusion Probabilistic Models (DDPMs) to generate training image pairs for corneal endothelial cell segmentation. Additionally, we introduce the Fourier Residual Block with Frequency Selection (FRB-FS), which enhances image quality by targeting high-frequency components crucial for detailed textures. Our framework requires only a limited number of labeled corneal endothelial cell images for training while efficiently generating an unlimited number of high-resolution, high-quality image pairs from noise. This approach significantly reduces the dependence on large annotated datasets. Inspired by recent advancements in image generation, particularly the work of [19], who utilized DDPMs for retinal vessel segmentation, we extend the application of DDPMs to corneal endothelial cell segmentation.

Our contributions are as follows: (1) We propose a novel two-stage Denoising Diffusion Probabilistic Model (DDPM) framework for generating paired synthetic training data for corneal endothelial cell segmentation. (2) We introduce the Fourier Residual Block with Frequency Selection (FRB-FS), which enhances high-frequency details and reduces irrelevant low-frequency components, leading to improved image quality. (3) We validate our approach through extensive experiments on two benchmark datasets, demonstrating its superior performance over existing methods.

II. METHODOLOGY

Overview. The proposed framework generates corneal endothelial cell image pairs through a two-stage process: label synthesis and image generation. In the first stage, synthetic labels are generated from random samples drawn from a standard normal distribution. These labels guide the generation

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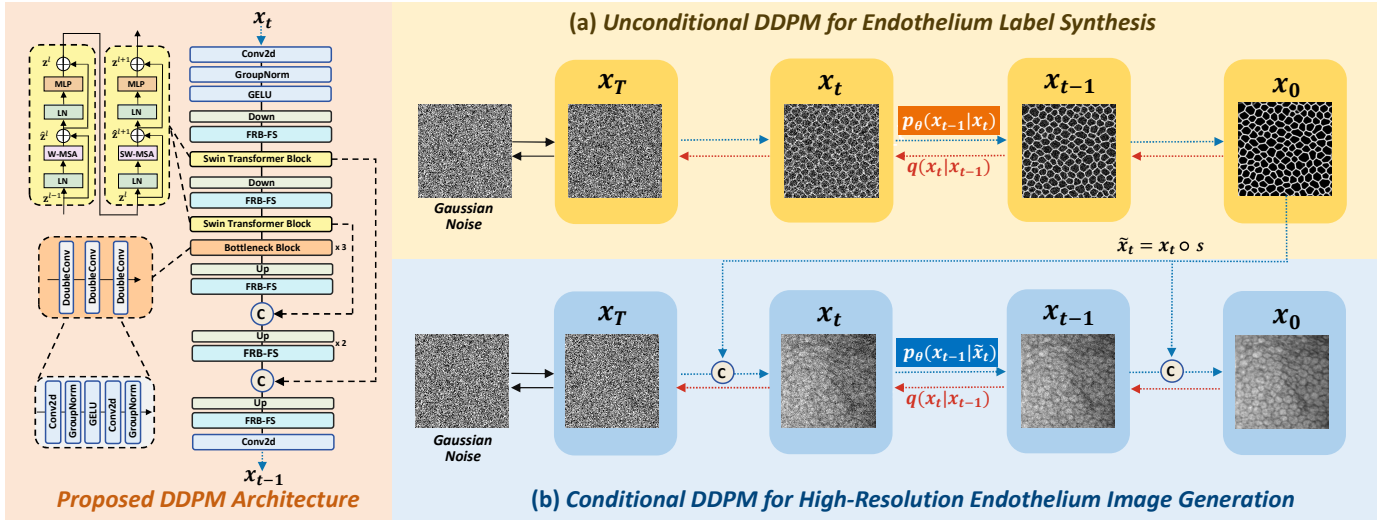


Fig. 1: Overview of our framework: (a) Unconditional DDPM for synthesizing endothelial cell labels, and (b) Conditional DDPM for generating high-resolution endothelial cell images.

of high-resolution images in the second stage, producing realistic, high-quality image pairs for training, as shown in Fig. 1.

A. Corneal Endothelium Label Synthesis

In corneal endothelium label synthesis stage, the model performs unconditional DDPM to generate synthetic endothelial cell labels, denoted as s . This process involves two main phases: the forward process and the reverse process.

During the forward process, Gaussian noise is gradually added to the initial label image x_0 over a series of diffusion time steps T . At each step t , the image x_t depends only on the previous step x_{t-1} , making this a Markovian chain. The noise is added using the variance schedule β_t , defined as:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}) \quad (1)$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (2)$$

where $\mathcal{N}(\cdot; \cdot, \cdot)$ represents a Gaussian distribution.

In the reverse process, the model gradually denoises the noisy image x_T using the neural network $p_\theta(x_{t-1}|x_t)$ to approximate the reverse conditional distributions:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (3)$$

To enhance the efficiency of the reverse diffusion process and accelerate convergence, we employ a cosine noise scheduler [20] instead of the linear scheduler used in the original DDPM work [21], resulting in faster convergence. The final step to sample an image from complete noise input x_T is:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right) + \sqrt{\beta_t} z \quad (4)$$

where $\epsilon_\theta(x_t, t)$ is the predicted noise, and if $t \neq 1$, then $z \sim \mathcal{N}(0, I)$; otherwise, z is 0. This process repeats from $t = T$

to $t = 1$ to obtain x_0 .

B. High-Resolution Corneal Endothelium Image Generation

In the generation of high-resolution corneal endothelium images, the DDPM process relies on previously synthesized endothelium labels. The conditional DDPM architecture concatenates the synthesized label with Gaussian noise at each step, guiding the image generation process. The process begins with the synthesized corneal endothelial label s , which is concatenated with the Gaussian noise input x_t at each diffusion step t , denoted as $\tilde{x}_t = x_t \circ s$. The reverse process in the conditional DDPM then progressively denoises this concatenated input $\tilde{x}_t$ to generate the high-resolution image. The denoising step is defined by $p_\theta(x_{t-1}|\tilde{x}_t)$. This procedure generates high-resolution images along with corresponding labels, thereby creating a substantial set of synthetic data for further training and evaluation.

C. The Architecture of DDPM

As shown in Fig. 1, the model begins with the input image x_t , which undergoes feature extraction through a 2D convolutional layer, followed by normalization and GELU activation. Features are then downsampled and processed through a Swin Transformer block [22] using Window Multi-head Self-Attention (W-MSA). After another downsampling and Swin Transformer block, the model enters a bottleneck phase where three convolutional blocks refine the features. During upsampling, two important concatenation operations occur: first, the input image x_t is concatenated with the label along the channel dimension during the forward pass; second, skip connections between the downsampling and upsampling stages are effectively leveraged by concatenating the skip tensor with the upsampled features along the channel dimension, ensuring the preservation of critical feature details. Additionally, Fourier Residual Blocks with Frequency Selection (FRB-FS) are integrated at each upsampling and downsampling stage to improve

the model's ability to process fine textures through frequency domain operations, thereby enhancing image reconstruction. The model concludes by passing the upsampled features through a final 2D convolutional layer to generate the denoised output x_{t-1} .

The Swin Transformer block uses Window Multi-head Self-Attention (W-MSA) to compute self-attention within fixed windows, reducing complexity from quadratic to linear:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \quad (5)$$

where Q , K , and V are the *query*, *key*, and *value* matrices, d is the *query/key* dimension, and B is the relative position bias.

D. Fourier Residual Block with Frequency Selection

We introduce the Fourier Residual Block with Frequency Selection (FRB-FS), which enhances high-frequency components related to blurred edges and fine textures while suppressing irrelevant low-frequency components through frequency domain operations. This dual function enables FRB-FS to generate denser, more accurate labels during DDPM sampling, thereby enhancing cell structure representation, as illustrated in Fig. 2. The process starts by applying a 2D Fourier Transform [23] to the input feature map $\mathbf{Z} \in \mathbb{R}^{H \times W \times C}$:

$$\mathbf{Z}_{\text{fft}} = \mathcal{F}(\mathbf{Z}) \quad (6)$$

where $\mathcal{F}$ denotes the Fourier Transform. Subsequently, a ReLU activation is applied to both the real and imaginary parts of $\mathbf{Z}_{\text{fft}}$, which emphasizes useful high frequencies and reduces low-frequency noise:

$$\mathbf{Z}_{\text{real}} = \text{ReLU}(\text{Re}(\mathbf{Z}_{\text{fft}})), \quad \mathbf{Z}_{\text{imag}} = \text{ReLU}(\text{Im}(\mathbf{Z}_{\text{fft}})) \quad (7)$$

Following 1×1 convolutions, these components are recombined using the inverse Fourier Transform:

$$\mathbf{Z}_{\text{conv}} = \mathcal{F}^{-1}(\text{conv}(\mathbf{Z}_{\text{real}}) + i \cdot \text{conv}(\mathbf{Z}_{\text{imag}})) \quad (8)$$

where $\mathcal{F}^{-1}$ represents the inverse Fourier Transform.

The process concludes by refining the real part of $\mathbf{Z}_{\text{conv}}$ through a spatial convolution and integrating a residual connection to generate the output $\mathbf{Y}$:

$$\mathbf{Y} = \text{Conv}_{\text{spatial}}(\text{Re}(\mathbf{Z}_{\text{conv}})) + \mathbf{Z} \quad (9)$$

FRB-FS not only enhances boundary clarity and contrast by selectively emphasizing high-frequency details but also facilitates faster model convergence by inheriting the advantages of ResBlock [24].

E. Loss Function

We use the combined loss function for parameter optimization θ in the denoising network $\epsilon_{\theta}(\cdot)$, aimed at predicting noise, can incorporate both the L_1 [25] and MSE loss metrics. The combined loss function is defined as follows:

$$\mathcal{L}_{DM} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\| + \|\epsilon - \epsilon_{\theta}(x_t, t)\|^2] \quad (10)$$

where x_t is computed as $\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$.

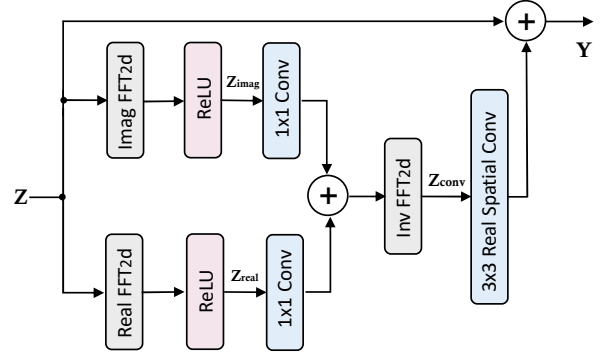


Fig. 2: Proposed Fourier Residual Block with Frequency Selection (FRB-FS).

III. EXPERIMENTS

A. Datasets

In this study, we employ two publicly available datasets from [26]: the Baerveldt and UT-DSAEK datasets, comprising a total of 1,203 corneal endothelial cell images acquired using a Topcon SP-1P microscope. These datasets cover both central and peripheral corneal regions, including cases with and without guttate. All images are 8-bit grayscale, with an original resolution of 240×528 pixels and have been manually annotated. For our experiments, we split the images into patches of size 256×256 pixels using nearest neighbour interpolation [27] to standardize the input for processing. Finally, we obtain 1,640 patches from the Baerveldt dataset and 766 patches from the UT-DSAEK dataset.

B. Implementation Details

All Experiments were conducted on three NVIDIA GeForce RTX 4090 Ti GPUs, each with 64 GB of memory. The U-Net [13] was trained for segmentation using Binary Cross-Entropy (BCE) Loss [28] to evaluate image quality. For the diffusion model, the learning rate started at 10^{-5} and followed a cosine decay schedule over 500 epochs. The Adam optimizer [29] was used for efficient weight updates. The noise schedule was managed across 800 diffusion steps to ensure precise denoising and image reconstruction during both training and inference.

C. Results

We assess the performance of our proposed framework by comparing it to two well-established GAN-based image synthesis models, Pix2Pix [30] and Patch-GAN [15]. First, we evaluate the quality of the generated images using the Fréchet Inception Distance (FID) score. As shown in Table I, our diffusion-based framework achieves the lowest FID scores of 53.42 and 57.04 on the respective datasets, significantly outperforming the GAN-based models by a substantial margin. To evaluate our method's effectiveness in corneal endothelial cell segmentation, we test it on the Baerveldt and UT-DSAEK datasets. We use 50% of the labeled data from each dataset to train our DDPM model, reserving the remaining 50% for performance evaluation. GAN-based methods, which cannot

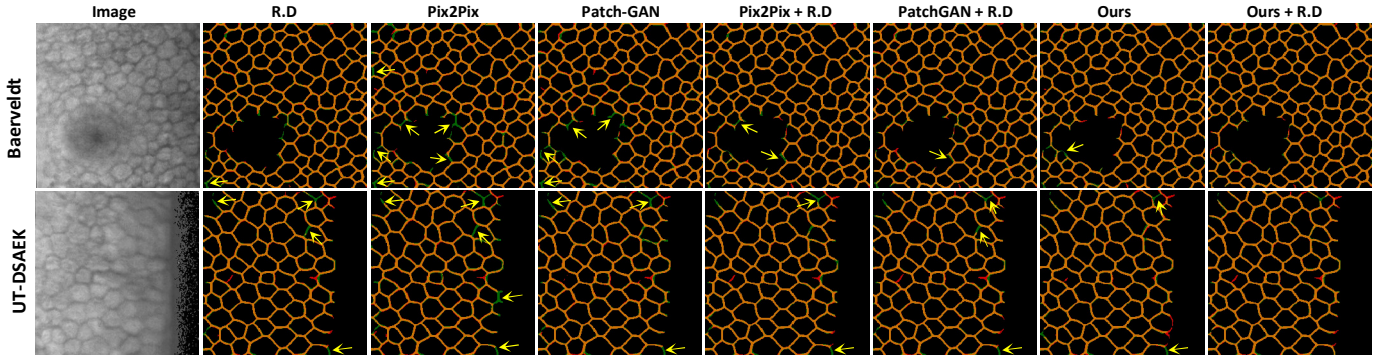


Fig. 3: Qualitative results on the Baerveldt and UT-DSAEK datasets are as follows: The prediction result is represented by a red line, while the ground truth is denoted by a green line, turning orange where they overlap. The yellow arrow indicates locations where the segmentation result does not align well with the ground truth.

TABLE I: Comparison of FID scores across different methods.

Method		FID Score ↓	
		Baerveldt	UT-DSAEK
GAN-based	Pix2Pix [30]	142.72	157.94
	Patch-GAN [15]	68.02	71.15
Diffusion-based	Ours	53.42	57.04

synthesize label images, rely on the first-stage synthetic endothelial labels generated by our DDPM. U-Net serves as the baseline segmentation model. For each method, we train two models: one with synthetic data only and the other with 50% raw data pairs (+R.D) used to train our DDPM. Segmentation performance is evaluated using the Dice Similarity Coefficient (Dice), Sensitivity (SE), Mean Accuracy (mAcc), and Modified Hausdorff Distance (MHD).

TABLE II: Comparison of U-Net [13] based corneal endothelial cell segmentation on benchmark datasets, using different training settings and performance metrics. The best and second-best results are in bold and underlined, respectively.

Test Data	Training Setting	Dice (%) ↑	SE (%) ↑	mAcc (%) ↑	MHD ↓
Baerveldt	Pix2Pix [30]	82.73 ± 1.98	80.16 ± 2.35	95.04 ± 1.77	0.582 ± 0.059
	Patch-GAN [15]	85.34 ± 2.13	83.22 ± 2.45	95.99 ± 1.84	0.435 ± 0.067
	Pix2Pix [30] + R.D	86.25 ± 1.82	83.95 ± 2.49	96.98 ± 1.79	0.294 ± 0.061
	Patch-GAN [15] + R.D	<u>87.17 ± 2.19</u>	<u>84.76 ± 2.21</u>	<u>97.56 ± 1.72</u>	0.195 ± 0.089
	Ours	86.68 ± 1.71	84.68 ± 2.35	97.63 ± 1.67	0.261 ± 0.084
	Ours + R.D	88.35 ± 1.91	85.52 ± 2.29	97.69 ± 1.86	0.132 ± 0.047
	R.D	86.66 ± 2.03	84.43 ± 2.42	97.35 ± 1.79	<u>0.148 ± 0.072</u>
UT-DSAEK	Pix2Pix [30]	81.25 ± 1.84	79.73 ± 2.46	94.91 ± 1.67	0.567 ± 0.069
	Patch-GAN [15]	84.37 ± 2.09	82.68 ± 2.31	95.19 ± 1.74	0.447 ± 0.063
	Pix2Pix [30] + R.D	85.28 ± 1.79	83.40 ± 2.48	96.05 ± 1.71	0.356 ± 0.078
	Patch-GAN [15] + R.D	86.46 ± 2.01	84.22 ± 2.41	96.78 ± 1.83	0.214 ± 0.087
	Ours	86.31 ± 1.93	83.90 ± 2.37	96.55 ± 1.68	0.284 ± 0.094
	Ours + R.D	87.24 ± 2.15	84.93 ± 2.29	97.49 ± 1.79	0.141 ± 0.053
	R.D	85.84 ± 2.04	83.76 ± 2.39	96.33 ± 1.72	0.162 ± 0.066

As shown in Table II, our proposed method, which combines synthetic data with raw data for training, sets a new benchmark by achieving state-of-the-art segmentation accuracy on both the Baerveldt and UT-DSAEK datasets. It significantly outperforms the baseline GAN models, including Pix2Pix [30] and Patch-GAN [15], across all metrics. Notably, our method achieves the highest Dice scores of 88.35% and 87.24%, respectively. Furthermore, our approach demonstrates superior performance with the highest mean accuracy (mAcc) and

the lowest Modified Hausdorff Distance (MHD), indicating more precise boundary delineation. Qualitative results in Fig. 3 illustrate that our method closely aligns with the ground truth, particularly in challenging regions where other methods struggle, as highlighted by the yellow arrows.

D. Ablation Study

We conduct an ablation study on both datasets to assess the impact of the FRB-FS module. As shown in Table III, adding the FRB-FS module significantly reduced the FID scores on the Baerveldt and UT-DSAEK datasets, indicating improved image quality and better alignment with real images. Additionally, the module decreased training time and accelerated model convergence.

TABLE III: Ablation study on the effectiveness of FRB-FS.

Dataset	FRB-FS	FID Score ↓	Training Time (hours) ↓
Baerveldt	×	67.26	19.53
	✓	53.42	18.41
UT-DSAEK	×	71.52	8.91
	✓	57.04	7.92

IV. CONCLUSIONS

In this study, we introduce a novel framework for generating corneal endothelial cell image pairs for segmentation tasks, utilizing DDPMs and the newly proposed FRB-FS module. Our method outperforms existing GAN-based models, demonstrating superior image quality and segmentation accuracy on the Baerveldt and UT-DSAEK datasets. The FRB-FS module further enhances image quality and reduces training time. Overall, our approach establishes a new benchmark in corneal endothelial image synthesis and segmentation.

V. ACKNOWLEDGEMENT

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